

Developmental Differences in Visual Processing: Strategy versus Structure

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The present study was an examination of two contrasting explanations for the finding that young children tend to classify objects according to similarity relations whereas adults emphasize dimensional structure. Subjects from three different age groups were given tasks that tapped different aspects of the visual process: Feature search, Conjunction search, and Restricted Classification tasks. In addition, three different stimulus objects (a highly separable, a highly integral, and an intermediate dimensional combination) were studied to explore the contribution of stimulus structure in dictating the nature of the processing mode. The results provided evidence against the view that children perceive objects according to a more primitive holistic structure. Rather, the results can be taken with past research to suggest that less efficient visual processing may lead the younger subjects to adopt a classification strategy that emphasizes similarity relations. In addition, the current research supported Garner's (1974, *The processing of information and structure*, Potomac, MD: Erlbaum) suggestion that the nature of the processing mode is stimulus-driven. © 1990 Academic Press, Inc.

The present study was an examination of children's performance on two different visual processing paradigms to distinguish between two competing theories of developmental differences in object perception. The study of the visual processing of multidimensional stimuli is interesting from a developmental perspective because different age groups have been found to respond in qualitatively different ways. When both children and adults are asked to group, rate, or sort multidimensional stimulus objects, 5- to 6-year-olds tend to make judgments based on overall similarity, whereas older children (11- to 12-year-olds) and adults tend to make judgments based on dimensional identity (Shepp, Burns,

This article is based on a dissertation conducted by the first author, under the supervision of the second author. We thank LeeAnne Flinn and Vickie Yeo for their assistance in data collection. We are also grateful to the students and teachers at Seeds University Elementary School. Reprint requests should be addressed to E. Ruskin at 68-237 Neuropsychiatric Institute, MRRC Department of Psychiatry, UCLA School of Medicine, 760 Westwood Plaza, Los Angeles, CA 90024-1759.

& McDonough, 1980; Shepp & Swartz, 1976; Smith & Kemler, 1977; Ward, 1980).

There are two major classes of theory to explain the observed developmental differences in object perception. The first class of theory (the "different-structure" view) presumes that there is a developmental change in perceived structure; that is, the younger subjects classify multidimensional stimuli on the basis of similarity relations because an integral stimulus structure is the most elemental stimulus organization, whereas the older subjects respond on the basis of dimensional identity because the separable stimulus structure is the most primitive stimulus organization. It is hypothesized that through perceptual learning and the development of abstract processes, a psychological reorganization of stimulus relations occurs, enabling the perceptual differentiation of a compound stimulus into its constituent dimensions. It should be noted that the strong version of the different-structure view (or the separability hypothesis, using Shepp's 1978 terminology) has been moderated on account of findings that young children are capable of analytic processing, although it is the least preferred processing mode.

The second class of theory (the "similar-structure" view), presumes that stimulus structure is perceived similarly by different age groups but that processing strategies change with age. Smith (1985) suggests that developmental differences may result from different classification decisions in response to certain demands; performance differences may reflect preferences for a particular type of classification, as well as variation in the flexibility to change from one classification strategy to another. The young child may be able to attend selectively in classification tasks but may not be motivated to classify by part identities (Evans & Smith, 1988).

Indeed, less efficient visual processing in children (Enns & Cameron, 1987) may lead the younger subjects to employ a classification strategy based on overall similarity relations. Smith and Kemler Nelson (1984) demonstrated that diminishing the processing efficiency of adults through speed stress and attentional capacity overload leads the adults to make a higher proportion of similarity than dimensional judgments in the classification task; that is, adults are induced to respond like children. To extend this observation, it is logical to suggest that children opt to use overall similarity relations to cope with their inefficient visual processing abilities.

A key study performed by Shepp, Barrett, and Kolbet (1987) demonstrated that the trend from holistic to analytic perception with increasing age is accompanied by an increase in attentional control. The results suggest that the developmental differences observed in classification and sorting tasks cannot simply be attributed to differences in

perceived stimulus structure, but also must account for the development of processing efficiency.

There are methodological problems with the evidence for both classes of theory. Treisman (1986) criticizes the paradigms used to support the different-structure view on the grounds that the tasks may tap into strategic processing rather than perceived stimulus structure. Using an alternative methodology (and an adult population), Treisman and her colleagues have shown that the feature is the most primitive unit of perception; the formation of the object is the result of selective attention to the separate features that have been analyzed at a preconscious and preattentive level (Treisman, 1986; Treisman & Gelade, 1980; Treisman & Paterson, 1984; Treisman & Souther, 1985). Treisman (1986) appeals for a distinction between subjective early experiences and objective early processes; she notes that although only 0.5 s is required for complete perceptual analysis, the speeded version of the restricted classification tasks usually requires 2 s (but see Smith & Kemler Nelson's (1984) Experiment 1 for one example of complete perceptual analysis under 2 s). Treisman suggests that feature and conjunction search may measure processes that occur temporally prior to those processes measured by classification paradigms. The paradigms used to demonstrate different perceived stimulus structures may instead be measuring different perceptual experiences.

Unfortunately, Treisman's conclusion of analytic primacy is based on too limited an array of stimulus dimensions. Garner (1974, 1976, 1978) and Garner and Felfoldy (1970) assert that different combinations of dimensions interact in different ways to produce two subcategories of objects: separable or integral (the "dimensional-interaction" hypothesis). A stimulus object is considered integral if the subject is unable to attend selectively to its separate components. Thus according to Garner, early perception is not necessarily analytic or holistic; rather the nature of the stimulus item dictates the processing mode. Similarity relations are not derived necessarily, but may be appreciated directly. As Treisman's criticisms on methodological issues have implications for the interpretation of the developmental differences observed on classification tasks it is necessary to test the analytic primacy theory both developmentally and with the dimensions deemed integral by other researchers.

In summary, the primary contention of this paper is that there is a confounding of theory with the paradigms and the stimulus dimensions used to study object perception in children. The purpose of the present experiment is to address this contention by providing two types of converging evidence in a single experiment. Task convergence will be examined by presenting three age groups with two different tasks that impose varying degrees of constraint on the nature of processing. To address the criticism that classification tasks measure strategy instead

of structure, within-subject performance on feature and conjunction search will be compared to performance on restricted classification.

Stimulus convergence will be examined by presenting subjects with three different types of dimensional combinations. To address the criticism that the analytic primacy view has not been tested with the dimensions deemed integral by other researchers, between-subject performance will be compared with an object that had been found to be highly separable on restricted classification tasks (a circle size and radius orientation combination; Ward, Foley, & Cole, 1986), an object that had been found to be highly integral on restricted classification tasks (a saturation and brightness combination; Garner & Felfoldy, 1970; Shepp & Swartz, 1976), and an object that had been found to elicit developmental differences in classification (a size and brightness combination, referred to as an "intermediate" stimulus object here; Smith & Kemler, 1977, 1978; Ward, 1983).

The observation of parallel or serial processing with the feature and conjunction targets serves to describe the nature of the stimulus representation. Separable encoding is distinguished by parallel processing in feature search and serial processing in conjunction search; integral encoding is demonstrated by parallel processing for both types of search (i.e., the integral nature of the stimulus dimensions leads subjects to treat the feature targets as conjunctions of features). The proportion of dimensional versus similarity judgments suggests separable versus integral encoding on the restricted classification task.

If the similar-structure view is supported, then the mode of processing (i.e., holistic or analytic) observed on the search tasks should be different from the mode of processing observed on the classification task (discontinuous processing). In particular, if there are qualitative developmental differences in performance on just the classification measure, and quantitative developmental differences (efficiency) on the two search tasks, then the view that developmental differences in object perception are due to different strategies rather than structure would be supported. If both measures tap the underlying stimulus structure, however, then the processing mode should be continuous across the different measures (at least for the younger subjects); this pattern of findings would provide converging evidence for the different-structure explanation of developmental differences in classification.

METHOD

Subjects

The participants in this experiment were 117 children who attended Seeds University Elementary School, a university-run school in the Los Angeles area. Three different age levels were tested: 5- and 6-year-olds ($M = 6;1$; range = 5;1 to 6;9), 7- and 8-year-olds ($M = 7;6$; range =

7:0 to 8:11), and 11- and 12-year-olds ($M = 11:7$; range = 10:10 to 12:6).¹ Each child in the two younger age groups was tested in three separate sessions. Each child in the oldest age group was tested in two separate sessions. The order of the three tasks was completely counterbalanced. There was an approximately equal number of boys and girls in each of the treatment groups.

Equipment

The three tasks were presented on an IBM 8514 color monitor, driven by an IBM PS/2 Model 50 with a 8514A memory-enhanced graphics adaptor, which allowed for the generation of 256,000 different colors.

General Procedure

The experiment was conducted in a partially darkened room, which was separated from the classroom. All subjects were given 5 min to dark adapt before the experiment was started. Task instructions were given during the adaptation time. The subjects were 55.88 cm from the computer screen for the duration of the experiment. Subjects were screened for visual acuity and color blindness; no subjects had to be eliminated from the study due to visual problems. Each subject participated in feature and conjunction search as well as nonstressed restricted classification. Each subject was assigned randomly to either the "Separable," "Integral," or "Intermediate" stimulus condition (13 subjects per condition for each age group). Thus a mixed design was used: 4 (Age) $\times$ 3 (Stimulus Type) $\times$ 3 (Task), with Age and Stimulus Type as between subjects variables, and Task as a within subjects variable.

The Search Tasks

Stimulus Materials

Although the search tasks were closely modeled after the search paradigms used by Treisman and Gelade (1980), display-size improvements suggested by Pashler (1987) were incorporated. Five display sizes of 2, 4, 8, 16, and 24 items were used in each condition. Each of these display sizes was presented in an outlined area subtending 10.16×10.16 cm ($10^\circ 25' \times 10^\circ 25'$).

For the two younger age groups, each search task consisted of 160 trials, divided into two major blocks of 80 trials each. Each block was subdivided into eight subblocks of 10 trials each. The two older age groups were given an extra block of 80 trials, for a total of 240 trials for

¹ In addition to the three groups of children, 39 college-aged adults who were enrolled in an introductory psychology course participated in the study. Analyses revealed few differences in processing efficiency and classification between the 11- and 12-year-olds and the adults, however, so the present paper will report the findings from the three youngest age groups.

each of the search tasks (eight extra subblocks). On half the trials the target was present, and on half the trials the target was absent. Each subblock contained a target-present and a target-absent trial for each of the five display sizes. The target positions were distributed evenly throughout the display area across trials. In the target-absent trials the distractors were arranged so that they matched the corresponding target-present trials, except that, of course, the target location was filled with an additional distractor item.

The distractors and targets for the different stimulus conditions were constructed in a similar fashion. As the logarithmic scale is most appropriate for creating psychologically equivalent stimulus intervals for the size dimension, the small, medium, and large sizes for the separable and the intermediate stimulus objects increased logarithmically. Past researchers have used color chips from the Munsell Color System as the saturation and brightness stimuli (as well as for the brightness dimension for the size and brightness object); the present experiment, however, presented the saturation and brightness stimuli on a computer. The computer colors that were used will be related to the corresponding Munsell color chip according to the Munsell Color System terminology. The translation of the reflective color chips into matching, self-illuminating colors on the computer screen was a complex process, details of which are available (Kaye & Ruskin, 1989; Ruskin, 1989). Table 1 lists the specific target and distractor objects for the search tasks. To make the task more appropriate for the younger age groups, only two disjunctive feature targets were used compared to the four disjunctive feature targets used by Treisman and Gelade (1980): One target matched half the distractors on one dimension; the other target matched half the distractors on the other dimension.

Procedure

The subjects were given the following instructions:

This is a game called "Find It!" As part of this game you will see a number of objects on the computer screen. Your job is to find this object (the experimenter points to the target on the screen). This object is called a target. Sometimes the target will be hiding among objects that look like these (the experimenter points to the distractors on the screen). These objects are called nontargets. Sometimes the target will not appear at all, and all you will see will be these nontargets.

If you find the target, then you should press the "yes" key as fast as you can (the experimenter points to a highlighted key on the board). If you do not see the target, then you should push the "no" key as fast as you can (the experimenter points to a highlighted key on the board). You can use your first and second fingers to do this (the experimenter shows them how to hold their fingers over the keys).

TABLE 1
TARGET AND DISTRACTOR OBJECTS FOR FEATURE AND CONJUNCTION SEARCH

Stimulus type	Distractors	Feature targets	Conjunction target
Separable ^a	Small-sized circle (0.20 sq cm or 31' diameter) with ra- dius 45° from horizontal	Medium-sized circle with radius 0° from the horizontal	Medium-sized circle with radius 45° from horizontal
	Medium-sized circle (.41 sq cm or 44' diameter) with ra- dius 90° from horizontal	Large-sized circle (.85 sq cm or 1°4' diameter) with ra- dius 45° from horizontal	
Integral ^b	Unsaturated, medium bright red Munsell 5R 5/2 (IBM RGB 120,50,70)	Slightly saturated, bright red Munsell 5R 6/6 (IBM RGB 185,70,100)	Slightly saturated, medium bright red Munsell 5R 5/6 (IBM RGB 155,50,75)
	Slightly saturated, dark red Munsell 5R 4/6 (IBM RGB 137,30,50)	Saturated, medium bright red Munsell 5R 5/10 (IBM RGB 185,35,50)	
Intermediate	Small-sized square (side length 0.5 cm or 31') with me- dium gray (IBM RGB 140,70,95)	Medium-sized square with light gray color (IBM RGB 200,95,130)	Medium-sized square with medium gray
	Medium-sized square (side length 0.72 cm or 44') with dark gray (IBM RGB 105,50,70)	Large-sized square (side length 1.04 sq cm or 1°4') with medium gray	

^a Circles were presented in white (IBM RGB [red, blue, green] 255,255,255). The radii extended from the center of the circles to the circle's circumference, and were bright green in color (IBM RGB 0,0,255).

^b Squares were 0.72 × 0.72 cm or 1°4' × 1°4'.

Although part of the game is to show how quickly you can tell if the target is there or not, it is also important to be careful to not make too many mistakes. If you hear a short, high beep (like this) then you will know that you got it right. "Correct" means that you found the target, or that you knew it wasn't hiding there. If you hear a long, low beep (like this), then you will know that you got it wrong. "Incorrect" means that you missed finding the target, or that you thought it was there when it wasn't.

The experimenter also explained the purpose of a fixation point (an asterisk) that appeared in the center of the computer screen at the onset of each trial for 1 s. In addition, the experimenter showed the subject how the conjunction target comprised components of both distractors, or how the feature target was either larger/redder/less red/darker/lighter, etc., than the corresponding distractions. (Examples were shown on the computer screen.) Following the presentation of the fixation point, there was a blank screen for 500 ms, then the display appeared until a keypress occurred. In addition to the audio feedback, if a subject made an incorrect response, a screen indicating the nature of the mistake was presented (e.g., "you missed the target," "the target was not there"). If the subject exceeded a 5-s limit, then the computer prompted the subject to "Try and go a little faster." Similarly, if a subject responded too quickly (i.e., under 150 ms), then the computer gave the message, "Slow down, you are hitting the keys too early!" In the last two cases, subjects had to repeat the trials. The experimenter initiated the next trial after the subject understood the type of error that had been made. In addition, each of the subblocks started with a review of the target and distractors. An intertrial interval of 2 s occurred prior to the presentation of the fixation point for the next trial.

The Restricted Classification Task

Stimulus Materials

The triads used in the restricted classification task were modeled after those used by Burns, Shepp, McDonough, and Weiner-Ehrlich (1978) and Ward et al. (1986). As with the search tasks, an attempt was made to devise the stimuli for the current study in such a manner that they could be related to past research; certain improvements in the stimulus items were included, however.²

To balance the number of stimulus levels across the different stimulus conditions, the separable and impressionable stimuli also were constructed with just four levels. The same sizes, orientations, saturations, and brightnesses were used as in the search tasks, with one additional level. Triads were devised so that each possible pairing between two of

² After working closely with the Munsell color system, we found it was difficult to distinguish among the chips in the two adjacent, highest saturation levels. In the Munsell system the adjacent chroma levels increment by two; a decision was made to use every other Munsell chroma level (that is, every fourth chroma level) to increase the discriminability of the saturation and brightness combinations. One luxury of using the computer to present the saturation and brightness stimuli (instead of color chips adhered to index cards) was that it was possible to extrapolate mathematically the IBM RGB numbers to create the hypothetical additional saturation and brightness combinations for the deeper saturations and lighter values.

the three objects could be classified as a dimensional, similarity, or haphazard response (see Fig. 1). To structure the triads so that they could be classified in such a manner, four levels of each dimension were combined in several different ways: Two of the items were identical on one dimension and different by two levels on the second dimension (the "dimensional" pair; "A" and "B" in Fig. 1). The third item differed from the other two items on both dimensions. One of the dimensions comprising the third item was different by one level from one of the dimensions of one of the first two items (the "similarity" pair; "A" and "C" in Figure 1). The second dimension comprising the third item was very different (by three levels) from the corresponding dimension on the second of the first two items (the "haphazard" pair; "B" and "C" in Fig. 1). A pilot study that involved similarity scaling indicated that the spacings used were appropriate for making the dimensional, similarity, and haphazard pairs for all three stimulus objects. There were 10 possible triads given the spacings described previously; one of these triads was discarded due to the results of a pilot oddity task (a discriminability task). The remaining nine triads were constructed such that there were identical pairs for each of the two component dimensions. The triads were presented in a row in the center of the computer screen. Three versions of each triad were devised so that the positions of the three items (and their pairings) were counterbalanced. Thus 54 judgments were made for each stimulus set.

For the separable stimulus, the circle size and radius orientation combination, the triads were constructed from circle size areas of 0.20 (31' diameter), 0.41 (44' diameter), 0.85 (1°4' diameter), and 1.77 (1°32' diameter) sq cm, and radii of 0, 45, 90, and 135° from the horizontal. The circles increased in area on a logarithmic scale. The radii and colors were the same as described for the search condition.

For the integral stimulus, the saturation and brightness combination,

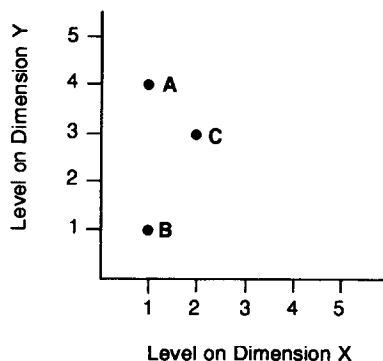


FIG. 1. Representation of a stimulus triad.

the triads were constructed from four levels of saturation ranging from unsaturated to highly saturated red, and four brightness levels ranging from very light to very dark. The computer generated saturation and brightness chips that corresponded to chips in the Munsell 5R hue chart; the IBM RGB (i.e., red, blue, and green) numbers are presented in Fig. 2.

For the intermediate stimulus, the size and brightness combination, the triads were constructed from squares that have side lengths of 0.5 (31'), 0.72 (44'), 1.04 (1°4'), and 1.5 (1°32') cm. The squares were filled with four gradients of gray ranging from light gray to a deep charcoal gray (IBM RGB 200,95,130; IBM RGB 140,70,95; IBM RGB 105,50,70; and IBM RGB 90,30,50, respectively). Again, the procedure for choosing these brightness levels was the same as that used to determine the brightness levels for the search tasks. The distance between each two of the three objects on the computer screen differed because the size of the three objects varied (at least for separable and intermediate stimuli). The interitem distance ranged between 1.1 cm for the two largest squares to 2.2 cm for the two smallest squares.

Procedure

The subjects were shown a practice triad and were told that the object of the game was to choose the two objects of the three that "go together the best." The experimenter explained that this means they could choose the first two objects, the last two objects, or the two objects on the outside (the experimenter pointed to the different combinations during the explanation). The subject was reminded intermittently of the purpose of the game throughout the experiment. The 11- and 12-year-olds re-

Value (brightness)	7/	200,85,122	210,77,108	227,62,93	240,50,80	E
			FT			E
	6/	174,75,105	185,70,100	212,55,85	235,35,55	
		D	CT	FT		
5/	120,50,70	165,50,75	185,35,50	200,20,35		
		D				
4/	113,40,60	137,30,50	167,0,0	190,0,0		
	/2	/6	/10	/14		
Chroma (saturation)						

E = Extrapolated Munsell Color Strips
FT = Feature Target
CT = Conjunction Target
D = Distractors

FIG. 2. IBM RGB (red, blue, green) numbers corresponding to Munsell Hue 5R for the search and classification stimulus objects with the saturation and brightness dimensional combination.

sponded by pressing two of the three keys that corresponded to the objects on the screen (e.g., the first key was used to indicate the choice of the first object, the second key was used to choose the middle object, and the third key was used to choose the third object; the three response keys were presented in a row to correspond to the row presentation of the stimuli). Some subjects in the two younger age groups had difficulty understanding the correspondence between the keys and the objects; in addition, some of these young subjects would inadvertently press one key, and then as they were pressing the second key they would inadvertently press the first key a second time. Thus some subjects in the younger age groups were told to point to their choice of the two objects that went together the best, and the experimenter entered their response. There were 12 practice trials. Subjects were encouraged to think about each response but to respond in less than 7 s. Subjects had to repeat any trial for which RT was shorter than 150 ms or longer than 7 s.

RESULTS

The Search Tasks

Reaction times (RTs) greater than 3.0 standard deviations from the mean of each respective combination of age, stimulus type, search, probe, and display size condition were treated as outliers and were eliminated from further analysis. The number of extreme scores, errors, and missing trials was computed for each subject (some subjects had several missing trials due to computer failure): Within each stimulus type and age group, the subject with the most missing data and extreme scores was identified and these data were discarded prior to the analysis. The mean median RTs for correct trials only for each age group, stimulus type, and search condition are shown in Figs. 3 and 4, which are discussed in subsequent sections. Overall, the error rates were quite low, ranging from 11% false negatives for the youngest subjects in conjunction

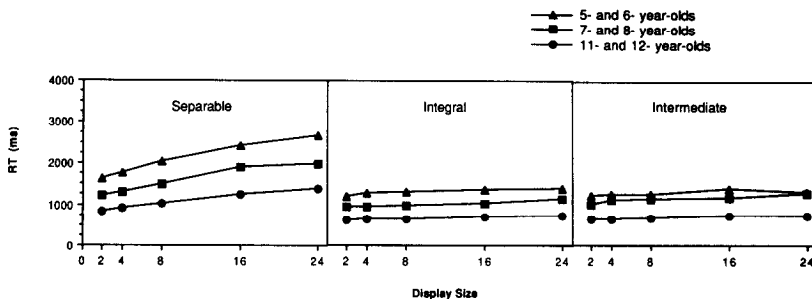


FIG. 3. Mean median correct RTs for conjunction search for all three age groups by stimulus type.

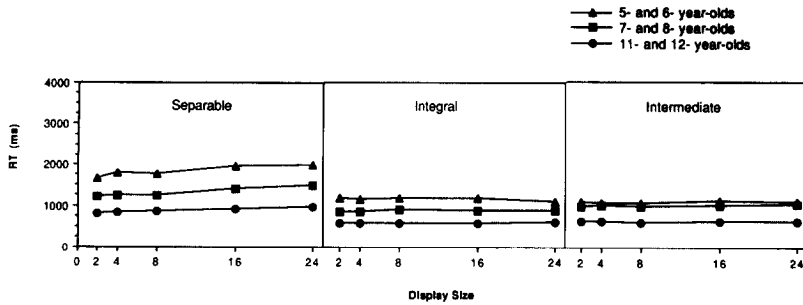


FIG. 4. Mean median correct RTs for feature search for all three age groups by stimulus type.

search for the separable stimulus to 2% each false positives and false negatives for the older subjects in feature search for integral stimulus.

An analysis of variance (ANOVA) was performed on the RT data. The analyses revealed both lower and higher order effects of condition, age, and stimulus type. In addition to significant main effects of condition, $F(1, 132) = 45.43, p < .001$; age, $F(3, 132) = 145.52, p < .001$; and stimulus type, $F(2, 132) = 109.12, p < .001$, a significant four-way interaction was found for Condition $\times$ Probe $\times$ Stimulus Type $\times$ Age, $F(6, 132) = 3.75, p < .01$. To compare the median RTs of the different age groups for each of the stimulus objects, additional t tests were computed. All median difference statements are based on two-tailed t tests with $\alpha = .05$. In addition, as performance on the negative probe condition is not the focus of the present article, reports will be limited to the positive probe condition. The median RTs for the separable stimulus object were significantly longer than the median RTs for either the integral or the intermediate stimulus objects for all three age groups and both feature and conjunction search. The median RTs for the integral stimulus object were not significantly different from the median RTs for the intermediate stimulus object among all three age groups. The results of this initial analysis suggest that the separable stimulus object is processed in a different manner from either the integral or intermediate stimulus objects. Furthermore, paired comparisons revealed that the median RTs for feature search were significantly shorter than the median RTs for conjunction search for all three age groups and stimulus types except for the 11- and 12-year-olds with the intermediate stimulus object.

To determine whether the differences in median RTs for the various stimulus and search conditions represented qualitative or quantitative differences in processing, the nature of processing was examined using a slope cutoff method. To review, if both feature and conjunction targets are processed in parallel, then integral encoding is suggested. In contrast, if feature targets are detected in parallel, but the identification of the

conjunction targets requires serial processing, then separable encoding is indicated. It was decided that a slope cutoff method was a more appropriate means of analysis than traditional measures of significance (i.e., deviation of a slope from zero in association with a high squared multiple correlation coefficient). The variance for the two younger age groups was much greater than for the oldest age group: Root mean square errors for the youngest age group were two to eight times higher than the root mean square error for the highly reliable oldest age group. (See Table 2 for the root mean square errors.) A slope of 20 ms/item was chosen as the cutoff between shallow slopes (parallel processing) and steep slopes (serial processing) based on the results of a cluster analysis and past research (Egeth, Virzi & Garbart, 1984; Murphy & Eriksen, 1987; Pashler, 1987; Treisman & Gelade, 1980). The cluster analysis and

TABLE 2

SLOPES, Y-INTERCEPTS (Y), F VALUES, R^2 'S, AND ROOT MEAN SQUARE ERRORS (RMSE) FOR THE POSITIVE PROBE CONDITION IN FEATURE AND CONJUNCTION SEARCH BY AGE GROUP AND STIMULUS TYPE

Stimulus type	Search condition	Regression statistics				
		Slope	<i>Y</i>	<i>F</i> value	<i>R</i> ²	RMSE
5- and 6-year-olds						
Separable	Feature	13.6	1671	18.5*	.86	57.5
	Conjunction	47.0	1588	138.9**	.98	72.8
Integral	Feature	−2.3	1183	1.8	.37	30.9
	Conjunction	7.3	1208	16.8*	.85	32.6
Intermediate	Feature	1.2	1077	1.0*	.26	22.3
	Conjunction	5.2	1206	2.6	.46	58.9
7- and 8-year-olds						
Separable	Feature	13.2	1171	74.6**	.96	27.9
	Conjunction	36.6	1175	42.8*	.93	102.1
Integral	Feature	1.3	855	1.4	.32	20.1
	Conjunction	8.3	906	44.2**	.94	22.8
Intermediate	Feature	2.2	983	9.4	.76	13.1
	Conjunction	9.6	1019	16.2*	.85	42.8
11- and 12-year-olds						
Separable	Feature	7.2	790	382.1**	.99	6.7
	Conjunction	24.8	800	183.6**	.99	33.4
Integral	Feature	0.6	581	2.7	.48	6.9
	Conjunction	5.0	600	71.0**	.96	10.7
Intermediate	Feature	0.4	624	0.5	.15	9.9
	Conjunction	4.2	628	36.8**	.92	12.4

Note. Slope units are milliseconds/item. Intercept values are in milliseconds.

* $p < .05$.

** $p < .01$.

the literature are discussed in detail by Ruskin (1989). The performance on the search tasks failed to support the different-structure view but were consistent with the predictions of the similar-structure as well as Garner's dimensional-interaction hypothesis: *Within* a particular stimulus condition, all three age groups processed the stimulus object in a qualitatively similar manner; however, there were qualitative differences *among* the different stimulus objects.

The separable stimulus object was processed analytically by all three age groups. All three age groups had steep slopes (slopes exceeding the 20 ms/item cutoff) for the target-present condition of conjunction search for the separable stimulus object. The slopes for both search conditions by age group and stimulus type are shown in Table 2. The steep slopes were consistent with item-by-item (serial) search through the display to identify the conjunction target. Figure 3 presents the mean median RTs (for correct trials only) in the conjunction search condition by age group and stimulus type.

All three age groups had shallow slopes (slopes less than the 20 ms/item cutoff) for the target-present condition of feature search for the separable stimulus object. The shallow slopes confirmed the hypothesis that the subjects were able to detect the feature target using parallel instead of serial search. Figure 4 presents the mean median RTs (for correct trials only) in the feature search condition by age group and stimulus type.

The integral and intermediate stimulus objects were processed holistically by all three age groups. All three age groups had shallow slopes (slopes less than the 20 ms/item cutoff) for the target-present condition of both feature and conjunction search for both the integral and intermediate stimulus objects. The shallow slopes confirmed the hypothesis that the subjects were able to detect even the conjunction targets using parallel instead of serial search. While the exact cutoff for the minimum amount of time needed to shift and refocus attention has not been established in the literature, the culmination of past research indicates that it is unlikely that item-by-item search can occur as rapidly as 10 ms/item (Ruskin, 1989). Thus although the slopes for the conjunction target were steeper than the slopes for the feature targets, the size of the slopes suggests that even though both types of targets were processed holistically, there were differences in the discriminability between the two types of targets.³

In summary, the observation of different types of encoding for the

³ It is possible that the slightly steeper slopes observed for the conjunction targets reflect differences in discriminability between conjunction targets and disjunction feature targets. The conjunction targets shares features with the distractors on both dimensions and the disjunction targets share a feature with the distractors on just one dimension. Again, although the slopes are steeper for the conjunction than feature search, the subjects were able to process both types of targets in parallel (holistically).

three stimulus objects confirmed the prediction of the dimensional-interaction hypothesis: The nature of the stimulus object influenced the mode of processing (integral versus separable) adopted by subjects for the search tasks. The lack of qualitative processing differences on the search tasks within each of the three different stimulus objects not only demonstrates that the three age groups perceived similar stimulus structures, but also suggests that the differences tapped by the ANOVA and *t* tests must be quantitative in nature.⁴

Developmental differences in processing efficiency. To review, younger subjects were expected to be less efficient than the older subjects in both the formulation and the execution of a response. The developmental prediction that subjects become more efficient at serial search with increasing age was supported by the data. All slope difference statements are based on one-tailed *t* tests of $\alpha = .05$. A comparison of the slopes for the target present condition of conjunction search for the separable stimulus revealed that the slope for the 5- and 6-year-olds was significantly steeper than the slope for the 7- and 8-year-olds, and the slope for the 7- and 8-year-olds was significantly steeper than the slope for the 11- and 12-year-olds. The slopes and *Y*-intercepts for the different age groups and stimulus types are presented in Table 2. The slopes might have been steeper for the younger subjects because they required more time to apply selective attention to each object (perhaps due to differences in the discriminability of the targets for the different age groups), or they may have searched less systematically through the display.

The data also supported the developmental prediction that the younger subjects were less efficient than the older subjects at parallel search. A comparison of the slopes and *Y*-intercepts in feature search with the separable stimulus and in feature and conjunction search for the integral and intermediate stimulus dimensions, confirmed this developmental prediction. Although well below the 20 ms/item cutoff, the slopes were steeper for younger than older subjects during parallel search. In most cases (except for feature search with the integral stimulus object), the 7- and 8-year-olds had significantly steeper slopes than the 11- and 12-year-olds. The observation of higher *Y*-intercepts for the two younger

⁴ The results of the negative trials were omitted as they do not add to the overall findings. With one exception (the 7- and 8-year-olds with the intermediate conjunction targets), subjects processed the negative trials serially for the separable stimulus object and in parallel for the integral and intermediate stimulus objects. The slope ratios we observed with the separable stimulus object are similar to those observed by Treisman and Gelade (1980). (In Treisman and Gelade's experiment, a 2:1 negative-to-positive slope ratio was believed to be indicative of self-terminating serial search, whereas a greater than 2:1 ratio suggested that subjects scanned the negative probe displays without actually checking each item.) Given that both the positive and negative probe trials were processed in parallel with the integral and intermediate stimulus objects, however, the ratios cannot be interpreted in the same manner.

age groups suggested that the younger children took longer to initiate a response (perhaps due to a more rigorous response criterion), and/or were less adept at carrying out the motor component of the task. More time elapsed before the target "popped-out" of the display for the younger children. (As was found with the regression analysis, differences in variability may have masked differences between the slopes of the 5- and 6-year-olds and the other two age groups.)

Restricted Classification Task

The proportions of dimensional, similarity, and haphazard responses made by the different age groups for the three types of stimulus objects are presented in Table 3. As the classification responses were correlated, several χ^2 tests were computed to determine if the number of subjects that responded analytically, holistically, or haphazardly was independent of the types of stimulus objects or the age of the subjects. The z distribution was used to establish the confidence limits ($\alpha = .05$) around the chance probability of making a dimensional, similarity, or haphazard response across all three age groups and three stimulus types. As separate limits were set up for each type of response, it was possible for a subject to be classified as responding analytically, holistically, both analytically and holistically, or haphazardly.⁵

As predicted by Garner's dimensional-interaction hypothesis, the frequency of the various classification responses observed for the three different stimulus objects differed significantly from the expected frequencies, $\chi^2(6) = 37.17, p < .001$ for the 5- and 6-year-olds; $\chi^2(6) = 27.33, p < .001$ for the 7- and 8-year-olds; and $\chi^2(6) = 38.52, p < .001$ for the 11- and 12-year-olds. These differences are not so notable for the older subjects, as the nonstressed/nonspeeded version of the restricted classification task was studied; it would be expected that the lack of time or capacity constraints would allow the older subjects to dimensionalize all but the highly integral stimulus dimensions. The finding of differences for the youngest age groups, however, suggests that the stimulus object does play a role in driving the nature of the processing mode rather than processing being strictly holistic for the children.

In addition to confirming the hypothesis that the subjects would classify the three stimulus objects with different proportions of the various types of responses, a separate χ^2 test also provided evidence of age differences in the proportion of the various types of classification responses made by the subjects, $\chi^2(6) = 5.12, p < .001$ for the separable stimulus, and

⁵ Subjects that responded with a significant proportion of haphazard responses were never coded as being both haphazard and either dimensional or integral. Only 4 out of 144 subjects were categorized as haphazard responders; the haphazard responders were all in the two youngest age groups.

TABLE 3
PROPORTION OF DIMENSIONAL, SIMILARITY, AND HAPHAZARD RESPONSES MADE ON THE
RESTRICTED CLASSIFICATION TASK FOR EACH AGE GROUP AND STIMULUS TYPE

Stimulus type	Type of response		
	Dimensional	Similarity	Haphazard
5- and 6-year-olds			
Separable	.70	.22	.08
Integral	.23	.69	.08
Intermediate	.60	.33	.07
7- and 8-year-olds			
Separable	.67	.22	.11
Integral	.27	.66	.07
Intermediate	.56	.33	.11
11- and 12-year-olds			
Separable	.89	.08	.03
Integral	.23	.75	.03
Intermediate	.82	.16	.02

$\chi^2(6) = 17.20, p < .001$ for the intermediate stimulus object. Significance tests were performed to compare the different types of responses made by the three age groups (for the sake of brevity, all mean difference statements are based on two-tailed t tests of $\alpha = .05$, unless otherwise noted). The t tests revealed: First, there were no significant age differences between the two youngest age groups; second, the 11- and 12-year-olds had a significantly higher proportion of dimensional responses and a significantly lower proportion of similarity responses than either the 5- and 6-year-olds or the 7- and 8-year-olds for both the separable and the intermediate stimulus objects; and third, the younger subjects were more likely to use similarity relations to classify the intermediate stimulus object than to classify the separable stimulus object.

Although other researchers (Smith & Kemler, 1977) have found that the 7- and 8-year-olds tended to perform midway between the 5- and 6-year-olds and the 11- and 12-year-olds in terms of their classification choices, the present study did not find any differences in the types of classifications made by the two youngest age groups for any of the stimulus objects. The younger subjects failed to meet the prediction that they would classify the intermediate stimulus object holistically, as they did not make a greater proportion of similarity than dimensional responses. Despite the overall tendency to classify even the intermediate stimulus object dimensionally, it is important to note, however, that the two younger age groups did make significantly more similarity responses and significantly fewer dimensional responses than the oldest subjects. The younger subjects made twice the proportion of similarity judgments

made by the 11- and 12-year-olds for the intermediate stimulus object. Given these results, one can still ask whether the greater tendency of the two younger age groups to use similarity relations in classifying the intermediate and separable stimulus objects was due to differences in perceived structure or differences in classification strategy. No age differences had been expected for the integral stimulus, as the highly integral nature of the saturation and brightness dimensions are expected to induce even adults to respond in an integral fashion. In summary, the results of the restricted classification task generalize to the findings of other researchers, with one exception: In our sample, the younger children responded somewhat more dimensionally than would be expected from past research.

DISCUSSION

The present research extended past findings on perceived structure and development in several ways. The study provided support for the view that developmental differences on classification tasks are due to differences in strategy rather than perceived structure. The developmental differences typically observed on the classification task generally were replicated in the present study. When perceived structure was studied with search task performance, however, no qualitative developmental differences for feature and conjunction search were observed within each of the three stimulus objects studied. This lack of age-related differences in search task performance demonstrated that the three age groups perceived a similar stimulus structure for a particular stimulus object. These findings of similar search task performance are contrary to the different-structure view that young children perceive an integral stimulus structure whereas older children and adults perceive a separable stimulus structure. Thus it was not the presence of developmental differences in classification per se that was disputed in this study, but rather the interpretation of the classification task as an indicator of perceived stimulus structure.

As mentioned earlier, performance on the restricted classification task generally matched the developmental predictions. All three age groups classified the separable stimulus objects based on its individual dimensions of size and radius orientation.⁶ In contrast, all three age groups classified the integral stimulus object based on the overall similarity of

⁶ Shepp, Burns, and McDonough (1980) found size and radius orientation to be classified according to overall similarity relations by the youngest subjects. We believe that our circle size and radius orientation combination was psychologically more separable than in Shepp et al.'s (1980) study as we controlled for configural effects by allowing the radius line to extend from the center of the circle to its circumference; Shepp et al. (1980), on the other hand, controlled for the contribution of length of the orientation line as a third dimension by keeping the length of the orientation line constant for all circle sizes.

the saturation and brightness combinations. Although the two younger age groups classified the intermediate stimulus object based on its individual dimensions more frequently than its overall similarity relations, the younger age groups had a significantly lower proportion of dimensional responses and a significantly higher proportion of similarity responses than the 11- and 12-year-olds. Although our subjects were comparable in age to those studied in related research, the 5- and 6-year-olds in our study used dimensional identity to classify stimuli in the present study somewhat more frequently than had been observed in past research (Smith & Kemler, 1977, 1978; Ward, 1983). It is possible that the computer presentation of the stimuli led the younger subjects to perform at a more developmentally advanced level (similar to the 7- and 8-year-olds). Perhaps the computer presentation led the subjects to respond to a different aspect of the stimuli than when they were presented on separate index cards (as in past research).

Search task performance did not reveal any age-related differences in perceived stimulus structure but demonstrated substantial developmental differences in processing efficiency. It was found that size and brightness, as well as saturation and brightness, initially were encoded together: Subjects did not require attention to detect either the feature or the conjunction targets. Only the search performance with the size and orientation combination revealed that some dimensions initially were encoded separately: Subjects were able to detect the feature target preattentively, but they required an item-by-item application of attention to identify the conjunction target. Quantitative differences in search were observed for all three age groups across all three stimulus types. As evidenced by steeper slopes and higher *Y*-intercepts, the 5- and 6-year-olds and the 7- and 8-year-olds were found to be less efficient than the older subjects in the formulation and the execution of a response.

Thus the present research served to distinguish between the two competing theories of developmental differences in classification task performance. When performance on both search and classification was considered, it was apparent that the increased use of similarity relations by the younger subjects for the restricted classification task was not due to differences in perceived stimulus structure as predicted by the different-structure view. The results converge nicely with Shepp et al.'s (1987) finding that differences in perceived structure are accompanied by differences in resource allocation (or attentional efficiency). Although not tested directly by the present study, it appears that developmental differences in processing efficiency may have led the younger subjects to adopt a classification strategy different from that of the older subjects.

The finding of less efficient information processing by the younger subjects suggests that the age differences observed on the classification task might reflect a resource-saving strategy. As mentioned earlier, if

adults are given classification tasks that involve speed stress or capacity overload, then they also are induced to adopt a holistic processing mode in making classification decisions. The lenient time and capacity constraints of the present study allowed the efficient visual processors (the 11- and 12-year-olds and adults) to dimensionalize the stimulus objects in making their classification decisions. The tasks may have been more taxing, however, for the younger, less efficient subjects. If adults adopt a holistic processing strategy under conditions that demand attention, then it is logical to assume that children adopt a similar tactic to cope with their limited capacities.

In addition to demonstrating that children and adults perceive similar stimulus structures, the present study also provided support to Garner's (1974) hypothesis: The different types of processing observed in association with the various stimulus objects in the search task served to highlight the role of the stimulus object in determining the processing mode adopted by the subjects. In the case of the integral and intermediate stimuli, the whole object was found to be a more primitive perceptual unit than the dimensions comprising the object. The opposite was true for the separable stimuli: In this case the featural information was found to be the more elemental unit. Thus present study disconfirmed the analytic primacy conceptualization of visual processing and instead supported the dimensional-interaction hypothesis that different types of stimulus objects compel different modes of visual processing.

It is possible that the analytic primacy theory had been formulated on an insufficient variety of dimensional combinations. As discussed earlier, the analytic primacy theory had largely been based on the search paradigms used by Treisman and her colleagues. In most cases the stimuli studied by Treisman were separable stimuli, such as letters of different shapes and colors (Treisman & Gelade, 1980; Treisman, Sykes, & Gelade, 1977), circles and triangles of different colors and sizes (Treisman, 1986), and circles and lines (Treisman & Souther, 1985). It is interesting to note that the one case in which analytic (separable) encoding was observed in the present study was for the stimulus item that was most like the stimulus objects studied by Treisman.

Our results are similar to those reported by Shepp (1989). Shepp and his colleagues used a visual priming technique to demonstrate that RTs to conjunction targets are as fast to the feature targets for "easy-to-analyze integral" stimuli (similar to our intermediate) but are slower for spatially separate dimensions.

In his review chapter, Shepp attributes the observed differences between Treisman's finding of analytic primacy and his finding of holistic properties to the nature of the task instructions. (Treisman gave subjects a verbal description of the conjunction target whereas Shepp provided subjects with a visual prime.) Shepp (1989) demonstrated that the RTs

to find the conjunctions are shorter if subjects are given a visual prime and longer if they are presented with a named or auditorially presented prime.

In our study, subjects were presented with a visual example of the target at the beginning of each subblock (every eight trials) and after each error; otherwise the task was a Treisman-style paradigm. Although our findings with the intermediate stimulus object replicate Shepp's findings, the finding of analytic processing with the size and orientation combination (an object that was not spatially separate) supports Treisman's findings. We would like to suggest that it is not the nature of the task instructions that accounts for these differences, but rather there are subtle differences among stimulus dimensions that do not make them readily recognizable as either integral or separable; thus it is not spatial separation per se that leads certain stimulus dimensions to be encoded analytically. Furthermore, we would like to suggest that converging operations such as those used by us in the present experiment are necessary to determine the nature of the stimulus encoding.

Converging operations and the continuum of integrality. The present study not only supported the dimensional-interaction hypothesis but also corroborated the view that there is a continuum of integrality for different stimulus combinations. The general argument put forth by Barrett and Shepp (1988), Burns et al. (1978), and Garner (1974) is that different combinations of dimensions vary in the ease with which they can be analyzed; consequently, the degree of analyzability dictates the processing options that are available to the subjects. In addition to finding separable processing on the search tasks by all three age groups, the finding that even the youngest subjects classified the separable stimulus object with a high proportion of dimensional classification suggests that the circle-size and radius orientation combination represented the highly separable end of the continuum of integrality. The saturation and brightness combination was found to represent the highly integral end of the continuum of integrality; in this case, the dimensional combination did not permit selective attention to the individual components and the object was resistant to analysis, on both the search and the classification task—even for the oldest age group. The size and brightness combination was found to represent an object that was less separable than the circle-size and brightness combination (for the two younger age groups) and more separable than the saturation and brightness combination (for all three age groups). Although all age groups treated the size and brightness combination integrally on the task that required nonstrategic responding, the oldest subjects were able to gain access to the dimensional structure given the lenient task demands of the restricted classification task.

Without studying performance on the search tasks, differences between the highly separable stimulus object and the intermediate stimulus object

could be distinguished only in terms of the proportion of dimensional responses made by the different age groups on the classification tasks. In the case of the separable stimulus object, however, it appeared that selection was mandatory on the conjunction search task as serial search was required to conjoin the separate features. In the case of the intermediate stimulus object, subjects processed the combination of size and brightness in parallel on the search tasks, but they were able to attend selectively to the individual dimensions on the classification task. Thus the size and orientation combination was judged as more separable than the size and brightness combination based on the observation of serial versus parallel search on the search tasks.

As mentioned earlier, our study demonstrates that converging operations are required to determine the nature of encoding of the various stimulus objects. It should also be noted that the performance on the search paradigms cannot be used in isolation from the classification tasks to order the objects with respect to their level of integrality. As a case in point, in the present study, both the integral and the intermediate stimulus objects were processed integrally on the search tasks. Performance on the classification task, however, suggested that given sufficient time and resources, the intermediate stimulus was highly analyzable for the older subjects, although the integral stimulus was still highly resistant to analysis. Thus both tasks used together provided a more complete picture of an object's position along the continuum of integrality than did either task alone. In this case, performance on both tasks suggested that the integral stimulus object was more integral than the intermediate stimulus object with respect to the continuum of integrality. Thus the study supported Garner, Hake, and Eriksen's (1956) call for converging operations to delimit the properties of the perceptual process.

CONCLUSION

The research reported in this article has important implications for the study of developmental differences in perceived structure. The interpretation of the results suggests that if the classification paradigm is used to study perceived stimulus structure in children (or possibly a stressed restricted classification task to study adults) then the outcome is an adequate means of revealing perceived stimulus structure. The ability of the classification task to measure encoding, however, is dependent upon the choice of the appropriate task demands or subject sample. If capacity is not taxed sufficiently (or the subjects are too efficient) then it is possible that differences in strategy rather than structure are measured by classification performance. Given these risks, the search task is a more appropriate means to study stimulus structure—especially as the present study generalized the search paradigm to include integral stimulus dimensions. The study of performance on search and classification tasks

together can be used to etch an even more concise picture of stimulus encoding: The nature of the stimulus coding, group differences in processing efficiency, and the relationship between the different dimensional combinations and their respective places on the continuum of integrality are revealed through this combined methodology.

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RECEIVED: September 28, 1989; REVISED: January 24, 1990.